1. Business Understanding

Business Problem: In the telecommunications industry, customer churn is a critical challenge that leads to significant revenue loss. Predicting whether a customer will leave or stay helps telecom companies identify the key factors driving churn and take proactive measures to improve customer retention.

Business Objectives:

- Predict customer churn based on various features such as tenure, service type, payment method, etc.
- Identify high-risk customers (those likely to churn) so the company can take preventative action, such as offering promotions or improving customer service.
- Increase revenue by focusing retention efforts on high-risk customers.
- Improve customer satisfaction by targeting issues that lead to churn.

Key Questions:

- Which factors most influence customer churn?
- Can we develop a predictive model to classify customers as "likely to churn" or "unlikely to churn"?
- What proactive strategies can reduce churn?

Success Criteria:

- Improved prediction of churn.
- Actionable insights to reduce churn rate by at least 10%.

Data Understanding

Dataset Structure:

- **Total Records**: 7043 rows.
- **Total Columns**: 21 features.

Data Attributes & Types:

Numerical Attributes:

- tenure: Duration (in months) the customer has stayed with the company (Discrete numerical).

- MonthlyCharges: Amount charged to the customer monthly (Continuous numerical).
- TotalCharges: Total amount charged to the customer to date (Currently object “string”, needs conversion to numeric for analysis; Continuous numerical).

Categorical Attributes:

- **Binary Categorical Attributes:**
    - Gender: Customer’s gender (Male/Female)
    - SeniorCitizen: Determine the customer is a senior or not “65+ years old” (Yes=1/No=0; currently integer, should ideally be categorical)
    - Partner: Determine the customer has a partner or not (Yes/No)
    - Dependents: Determine the customer has dependents or not (Yes/No)
    - PhoneService: Determine the customer has a phone service or not (Yes/No)
    - PaperlessBilling: Determine the customer has paperless billing “electronically” or traditional (Yes/No)
    - Churn: The target variable, whether the customer has churned or not (Yes/No)
- **Nominal Categorical Attributes:**
    - MultipleLines: "No phone service means no phone plan at all, No means a single phone line, and Yes means multiple phone lines." (No phone service/No/Yes)
    - InternetService: Type of internet service (DSL, Fiber optic, No)
    - OnlineSecurity (No, Yes, No internet service)
    - OnlineBackup (No, Yes, No internet service)
    - DeviceProtection (No, Yes, No internet service)
    - TechSupport (No, Yes, No internet service)
    - StreamingTV (No, Yes, No internet service)
    - StreamingMovies (No, Yes, No internet service)
    - Contract :Type of contract (Month-to-month, One year, Two year)

    - PaymentMethod: Method of payment (Electronic check, Mailed check, Bank transfer, Credit card automatic)

- **Identifier Attribute**:
    - customerID: Unique identifier for each customer (categorical nominal, should not be used directly in modelling as it holds no predictive value).

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2. Relationships Between Attributes

1. **Churn and Contract Type**:
    - Customers on a **Month-to-month** contract are more likely to churn than those on **One year** or **Two year** contracts.
2. **Churn and Monthly Charges**:
    - Customers with **higher monthly charges** might be less likely to churn, but this is not always true. The relationship may be complex and dependent on other factors like the contract type or tenure.
3. **Churn and Tenure**:
    - There is likely a **negative relationship** between **Churn** and **Tenure**. Longer-tenured customers generally are less likely to churn because they have had a longer relationship with the company, and may be more accustomed to the services.
4. **Churn and Services (e.g., StreamingTV, TechSupport)**:
    - Customers with more services (such as **StreamingTV**, **TechSupport**, and **OnlineSecurity**) are generally **less likely to churn**. Customers who feel they are getting value from multiple services might stay longer.
5. **Payment Method and Churn**:
    - **PaymentMethod** could also show some relationship with **Churn**. For instance, customers using **Electronic checks** might have a higher churn rate compared to those using **Bank transfer** or **Credit cards**, but this needs further exploration.

6. **Churn and Customer Demographics (SeniorCitizen, Gender, Partner, Dependents)**:

    - Senior citizens may churn less, as they are more likely to have longer-term relationships with the company and may prefer stable services.
    - Gender and family situation (Partner/Dependents) could have minor impacts on churn but are secondary to other features like contract type and service features.
7. **Internet Service and Service Features**:
    - The availability of **InternetService** (DSL, Fiber optic, or None) is likely to influence whether customers opt for additional services such as **OnlineSecurity**, **TechSupport**, **StreamingTV**, and **StreamingMovies**. Customers with better internet connections (like **Fiber optic**) are more likely to subscribe to additional services.
    - Customers without internet service are less likely to have services like **TechSupport** and **StreamingTV**.

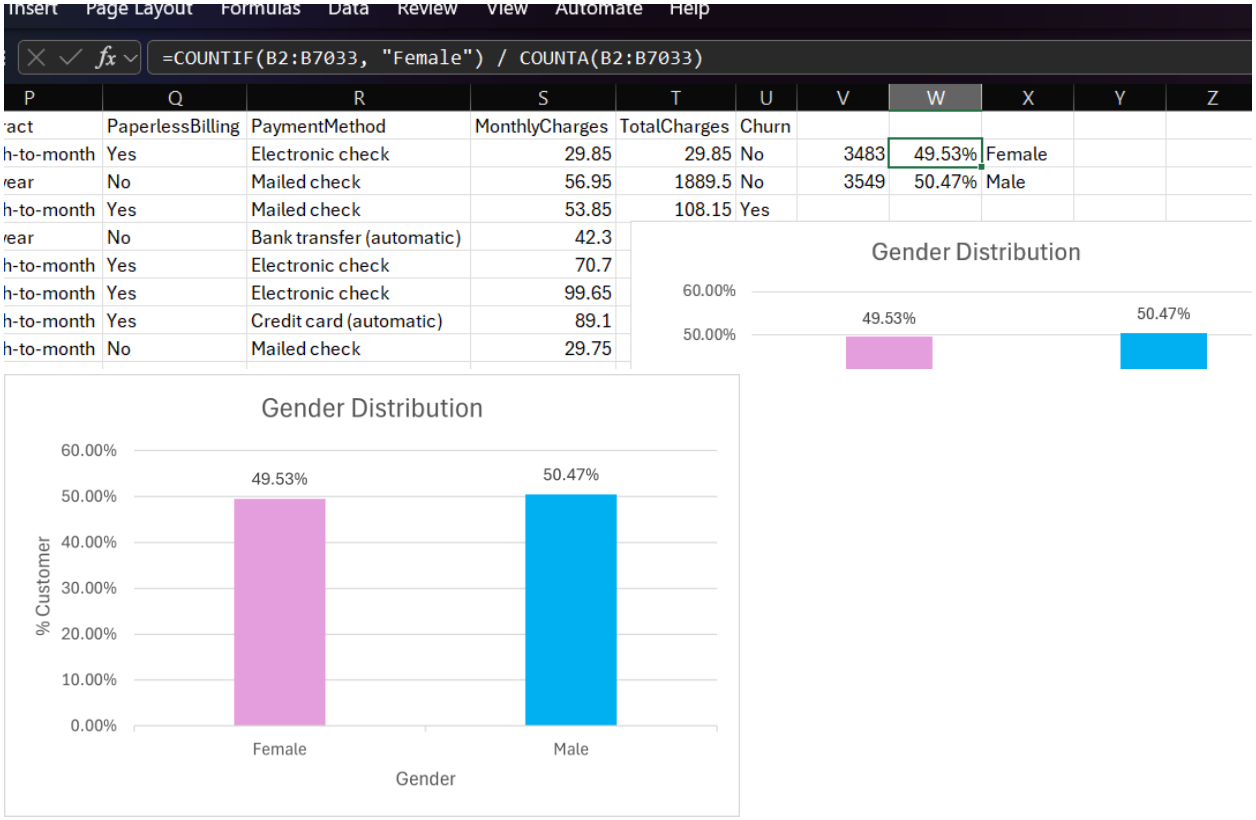
Data Quality Issues:

- **Missing or problematic values**:
    - TotalCharges is of type object instead of numeric, indicating potential issues such as empty strings that require conversion and cleaning.
- **Duplicates**:
    - customerID must be checked for duplicates to ensure unique identification.

A.) ***Demographics*** - Let us first understand the gender, age range, patner and dependent status of the customers

linkcode

1. **Gender Distribution** - About half of the customers in our data set are male while the other half are female

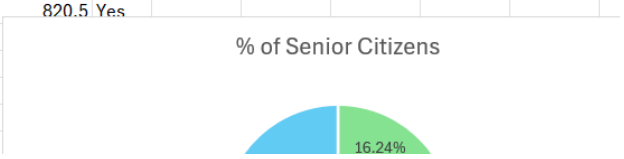


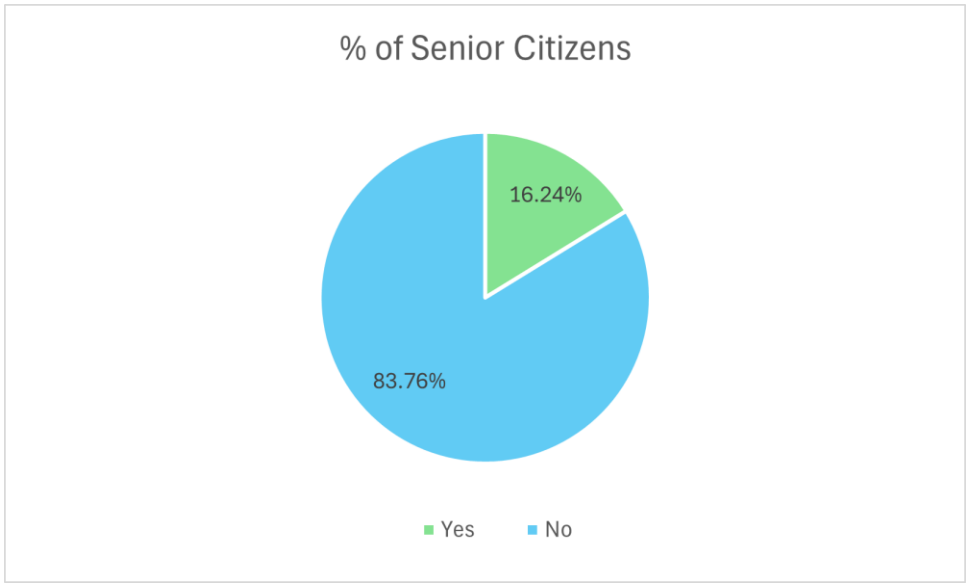
1. **% Senior Citizens** - There are only 16% of the customers who are senior citizens. Thus most of our customers in the data are younger people.

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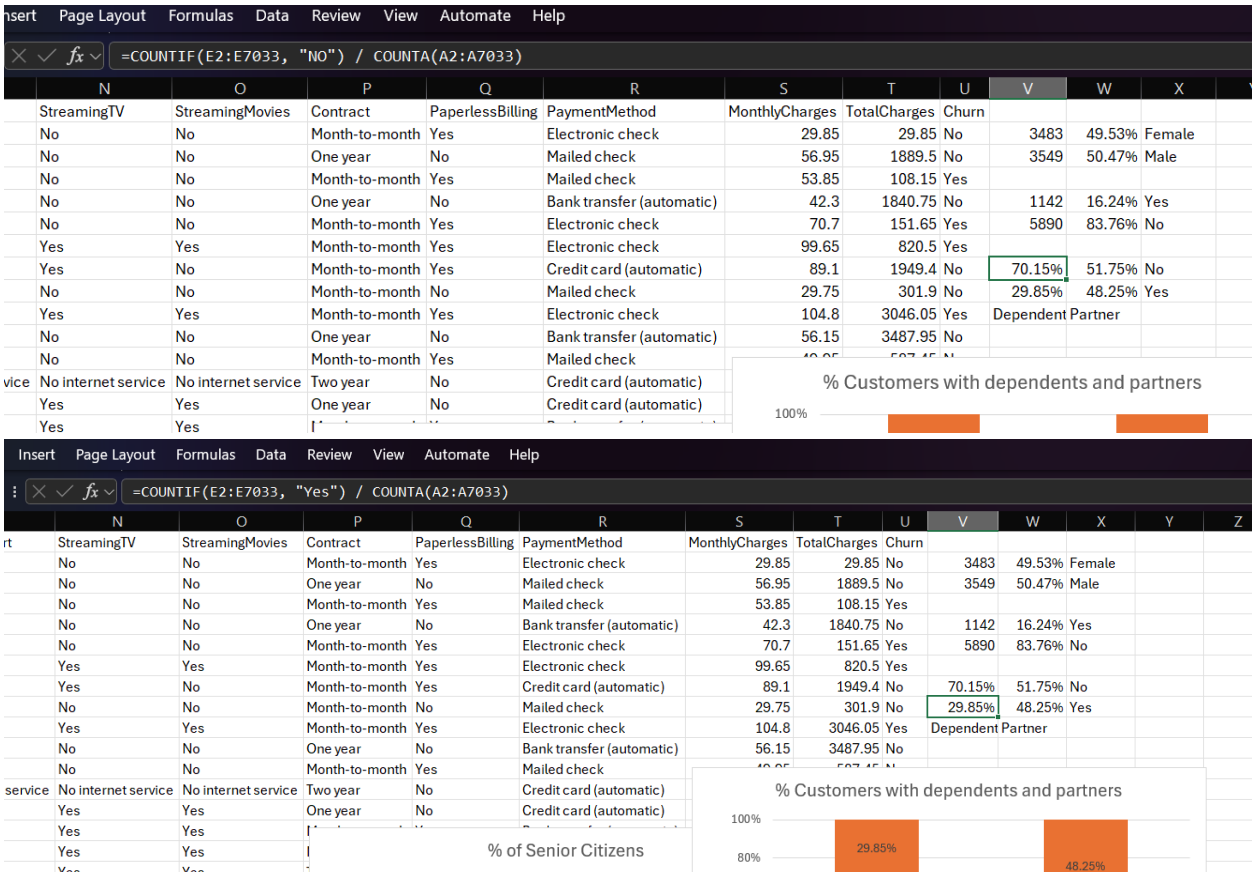
=COUNTIF(C2:C7033, "1") / COUNTA(C2:C7033)

P	Q	R	S	T	U	V	W	X	Y	Z
tract	PaperlessBilling	PaymentMethod	MonthlyCharges	TotalCharges	Churn					
ıth-to-month	Yes	Electronic check	29.85	29.85	No	3483	49.53%	Female		
year	No	Mailed check	56.95	1889.5	No	3549	50.47%	Male		
ıth-to-month	Yes	Mailed check	53.85	108.15	Yes					
year	No	Bank transfer (automatic)	42.3	1840.75	No	1142	16.24%	Yes		
ıth-to-month	Yes	Electronic check	70.7	151.65	Yes	5890	83.76%	No		
ıth-to-month	Yes	Electronic check	99.65	820.5	Yes					
ıth-to-month	Yes	Credit card (automatic)	89.1							
ıth-to-month	No	Mailed check	29.75							
ıth-to-month	Yes	Electronic check	104.8							
year	No	Bank transfer (automatic)	56.15							
ıth-to-month	Yes	Mailed check	49.95							





1. **Partner and dependent status** - About 50% of the customers have a partner, while only 30% of the total customers have dependents.



	StreamingTV	StreamingMovies	Contract	PaperlessBilling	PaymentMethod	MonthlyCharges	TotalCharges	Churn			
	No	No	Month-to-month	Yes	Electronic check	29.85	29.85	No	3483	49.53%	Female
	No	No	One year	No	Mailed check	56.95	1889.5	No	3549	50.47%	Male
	No	No	Month-to-month	Yes	Mailed check	53.85	108.15	Yes			
	No	No	One year	No	Bank transfer (automatic)	42.3	1840.75	No	1142	16.24%	Yes
	No	No	Month-to-month	Yes	Electronic check	70.7	151.65	Yes	5890	83.76%	No
	Yes	Yes	Month-to-month	Yes	Electronic check	99.65	820.5	Yes			
	Yes	No	Month-to-month	Yes	Credit card (automatic)	89.1	1949.4	No	70.15%	51.75%	No
	No	No	Month-to-month	No	Mailed check	29.75	301.9	No	29.85%	48.25%	Yes
	Yes	Yes	Month-to-month	Yes	Electronic check	104.8	3046.05	Yes	Dependent	Partner	
	No	No	One year	No	Bank transfer (automatic)	56.15	3487.95	No			
	No	No	Month-to-month	Yes	Mailed check						
vice	No internet service	No internet service	Two year	No	Credit card (automatic)						
	Yes	Yes	One year	No	Credit card (automatic)						
	Yes	Yes									

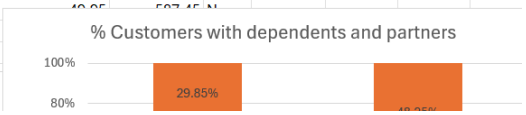
rt	StreamingTV	StreamingMovies	Contract	PaperlessBilling	PaymentMethod	MonthlyCharges	TotalCharges	Churn				
	No	No	Month-to-month	Yes	Electronic check	29.85	29.85	No	3483	49.53%	Female	
	No	No	One year	No	Mailed check	56.95	1889.5	No	3549	50.47%	Male	
	No	No	Month-to-month	Yes	Mailed check	53.85	108.15	Yes				
	No	No	One year	No	Bank transfer (automatic)	42.3	1840.75	No	1142	16.24%	Yes	
	No	No	Month-to-month	Yes	Electronic check	70.7	151.65	Yes	5890	83.76%	No	
	Yes	Yes	Month-to-month	Yes	Electronic check	99.65	820.5	Yes				
	Yes	No	Month-to-month	Yes	Credit card (automatic)	89.1	1949.4	No	70.15%	51.75%	No	
	No	No	Month-to-month	No	Mailed check	29.75	301.9	No	29.85%	48.25%	Yes	
	Yes	Yes	Month-to-month	Yes	Electronic check	104.8	3046.05	Yes	Dependent	Partner		
	No	No	One year	No	Bank transfer (automatic)	56.15	3487.95	No				
	No	No	Month-to-month	Yes	Mailed check							
service	No internet service	No internet service	Two year	No	Credit card (automatic)							
	Yes	Yes	One year	No	Credit card (automatic)							
	Yes	Yes										
	Yes	Yes										

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=COUNTIF(D2:D7033, "NO") / COUNTA(A2:A7033)

	N	O	P	Q	R	S	T	U	V	W	X	Y
rt	StreamingTV	StreamingMovies	Contract	PaperlessBilling	PaymentMethod	MonthlyCharges	TotalCharges	Churn				
	No	No	Month-to-month	Yes	Electronic check	29.85	29.85	No	3483	49.53%	Female	
	No	No	One year	No	Mailed check	56.95	1889.5	No	3549	50.47%	Male	
	No	No	Month-to-month	Yes	Mailed check	53.85	108.15	Yes				
	No	No	One year	No	Bank transfer (automatic)	42.3	1840.75	No	1142	16.24%	Yes	
	No	No	Month-to-month	Yes	Electronic check	70.7	151.65	Yes	5890	83.76%	No	
	Yes	Yes	Month-to-month	Yes	Electronic check	99.65	820.5	Yes				
	Yes	No	Month-to-month	Yes	Credit card (automatic)	89.1	1949.4	No	70.15%	51.75%	No	
	No	No	Month-to-month	No	Mailed check	29.75	301.9	No	29.85%	48.25%	Yes	
	Yes	Yes	Month-to-month	Yes	Electronic check	104.8	3046.05	Yes	Dependent	Partner		
	No	No	One year	No	Bank transfer (automatic)	56.15	3487.95	No				
	No	No	Month-to-month	Yes	Mailed check							
service	No internet service	No internet service	Two year	No	Credit card (automatic)							
	Yes	Yes	One year	No	Credit card (automatic)							
	Yes	Yes										
	Yes	Yes										

% of Senior Citizens

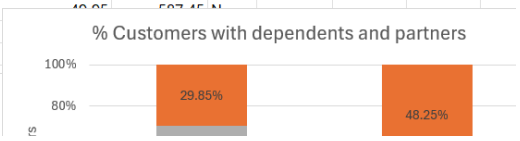


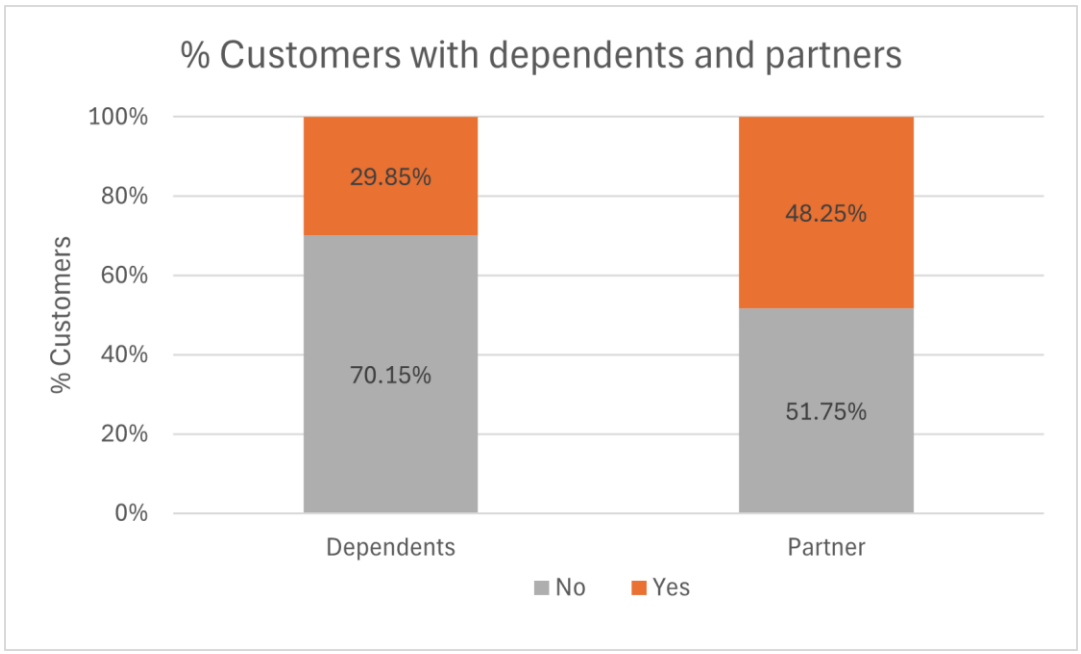
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=COUNTIF(D2:D7033, "Yes") / COUNTA(A2:A7033)

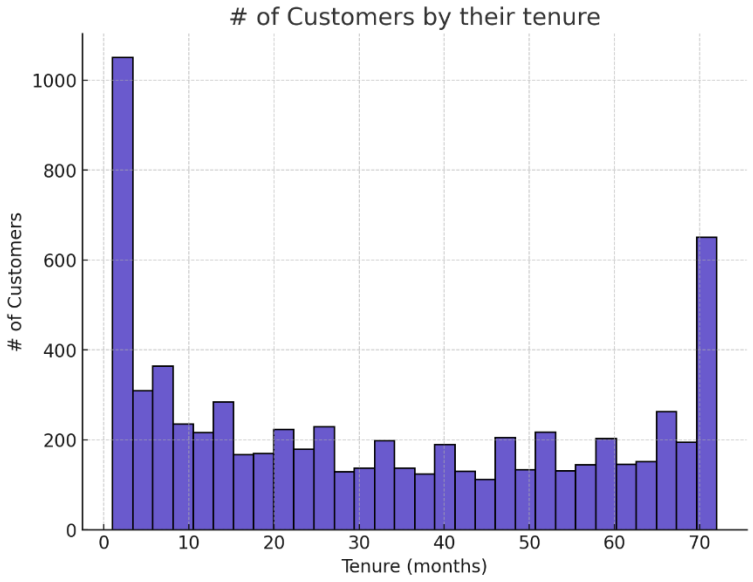
	N	O	P	Q	R	S	T	U	V	W	X
	StreamingTV	StreamingMovies	Contract	PaperlessBilling	PaymentMethod	MonthlyCharges	TotalCharges	Churn			
	No	No	Month-to-month	Yes	Electronic check	29.85	29.85	No	3483	49.53%	Female
	No	No	One year	No	Mailed check	56.95	1889.5	No	3549	50.47%	Male
	No	No	Month-to-month	Yes	Mailed check	53.85	108.15	Yes			
	No	No	One year	No	Bank transfer (automatic)	42.3	1840.75	No	1142	16.24%	Yes
	No	No	Month-to-month	Yes	Electronic check	70.7	151.65	Yes	5890	83.76%	No
	Yes	Yes	Month-to-month	Yes	Electronic check	99.65	820.5	Yes			
	Yes	No	Month-to-month	Yes	Credit card (automatic)	89.1	1949.4	No	70.15%	51.75%	No
	No	No	Month-to-month	No	Mailed check	29.75	301.9	No	29.85%	48.25%	Yes
	Yes	Yes	Month-to-month	Yes	Electronic check	104.8	3046.05	Yes	Dependent	Partner	
	No	No	One year	No	Bank transfer (automatic)	56.15	3487.95	No			
	No	No	Month-to-month	Yes	Mailed check						
ervice	No internet service	No internet service	Two year	No	Credit card (automatic)						
	Yes	Yes	One year	No	Credit card (automatic)						
	Yes	Yes									
	Yes	Yes									
	Yes	Yes									

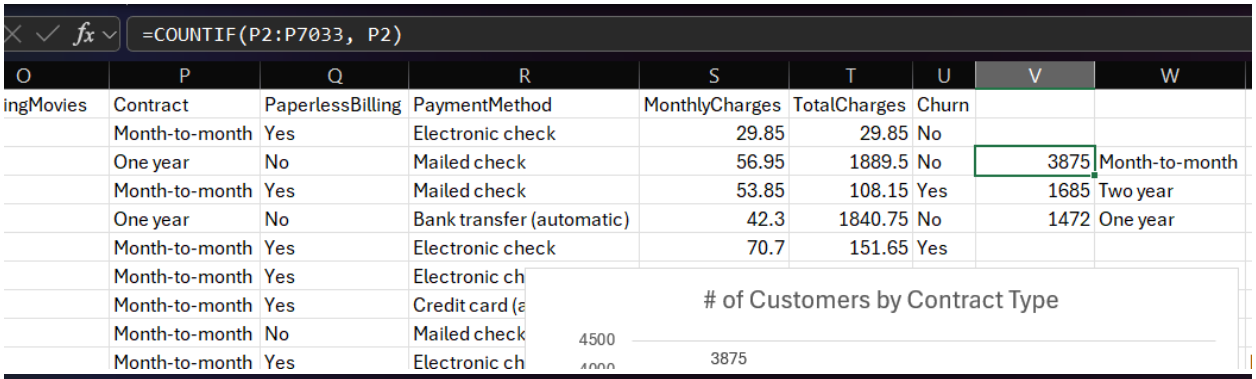
% of Senior Citizens

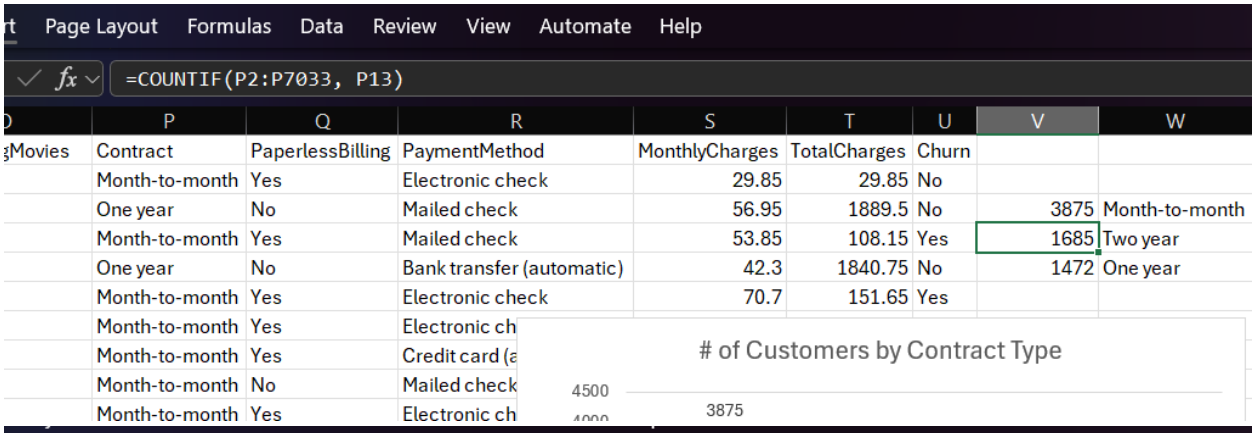


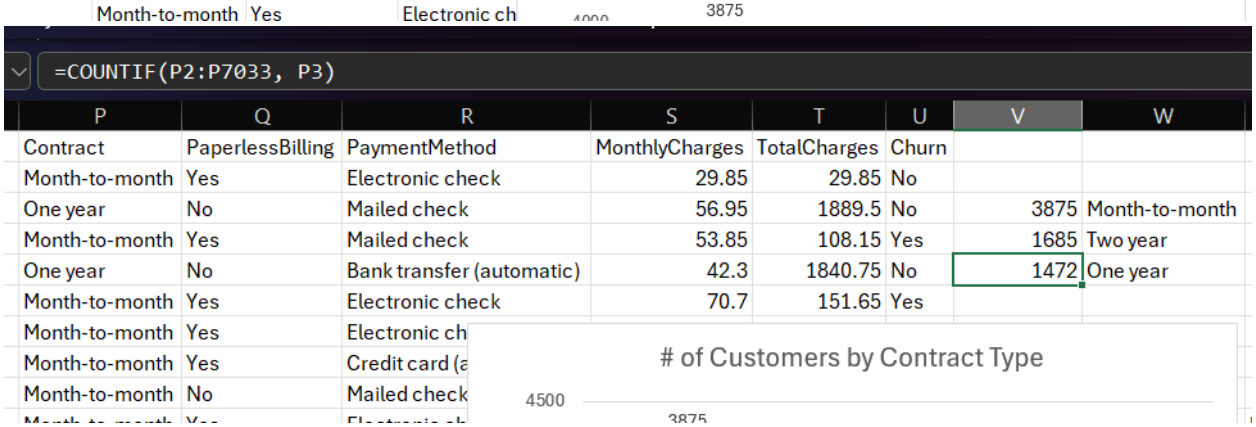


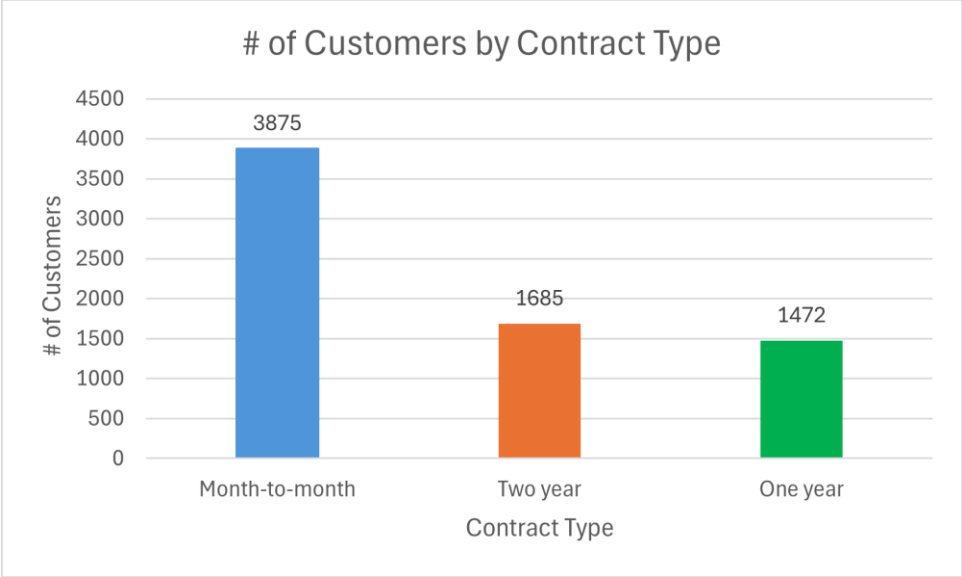
1. Tenure: After looking at the below histogram we can see that a lot of customers have been with the telecom company for just a month, while quite a many are there for about 72 months “6 years”. This could be potentially because different customers have different contracts. Thus based on the contract they are into it could be more/less easier for the customers to stay/leave the telecom company.



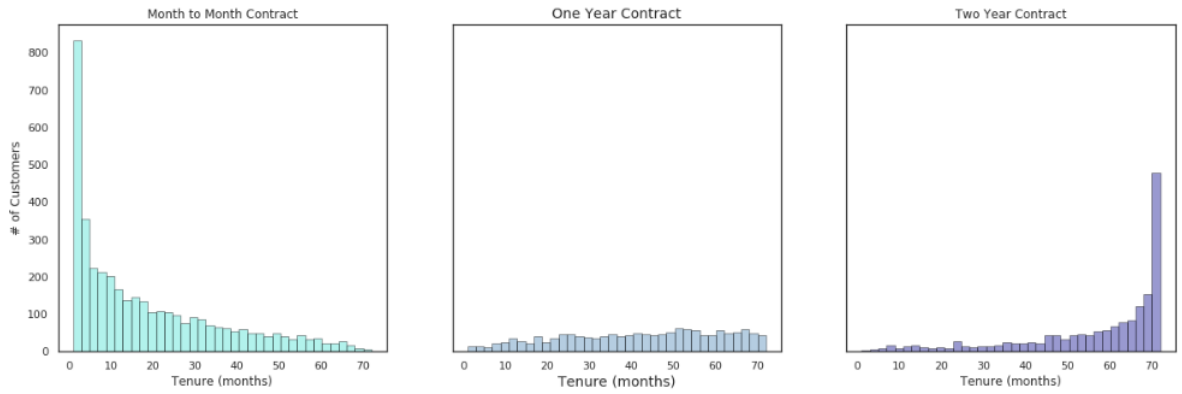

=COUNTIF(P2:P7033, P2)
ingMovies Contract PaperlessBilling PaymentMethod MonthlyCharges TotalCharges Churn
Month-to-month Yes Electronic check 29.85 29.85 No
One year No Mailed check 56.95 1889.5 No 3875 Month-to-month
Month-to-month Yes Mailed check 53.85 108.15 Yes 1685 Two year
One year No Bank transfer (automatic) 42.3 1840.75 No 1472 One year
Month-to-month Yes Electronic check 70.7 151.65 Yes
Month-to-month Yes Electronic ch
Month-to-month Yes Credit card (a
Month-to-month No Mailed check
Month-to-month Yes Electronic ch
of Customers by Contract Type
4500
3875


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=COUNTIF(P2:P7033, P13)
gMovies Contract PaperlessBilling PaymentMethod MonthlyCharges TotalCharges Churn
Month-to-month Yes Electronic check 29.85 29.85 No
One year No Mailed check 56.95 1889.5 No 3875 Month-to-month
Month-to-month Yes Mailed check 53.85 108.15 Yes 1685 Two year
One year No Bank transfer (automatic) 42.3 1840.75 No 1472 One year
Month-to-month Yes Electronic check 70.7 151.65 Yes
Month-to-month Yes Electronic ch
Month-to-month Yes Credit card (a
Month-to-month No Mailed check
Month-to-month Yes Electronic ch
of Customers by Contract Type
4500
3875


=COUNTIF(P2:P7033, P3)
Contract PaperlessBilling PaymentMethod MonthlyCharges TotalCharges Churn
Month-to-month Yes Electronic check 29.85 29.85 No
One year No Mailed check 56.95 1889.5 No 3875 Month-to-month
Month-to-month Yes Mailed check 53.85 108.15 Yes 1685 Two year
One year No Bank transfer (automatic) 42.3 1840.75 No 1472 One year
Month-to-month Yes Electronic check 70.7 151.65 Yes
Month-to-month Yes Electronic ch
Month-to-month Yes Credit card (a
Month-to-month No Mailed check
of Customers by Contract Type
4500
3875

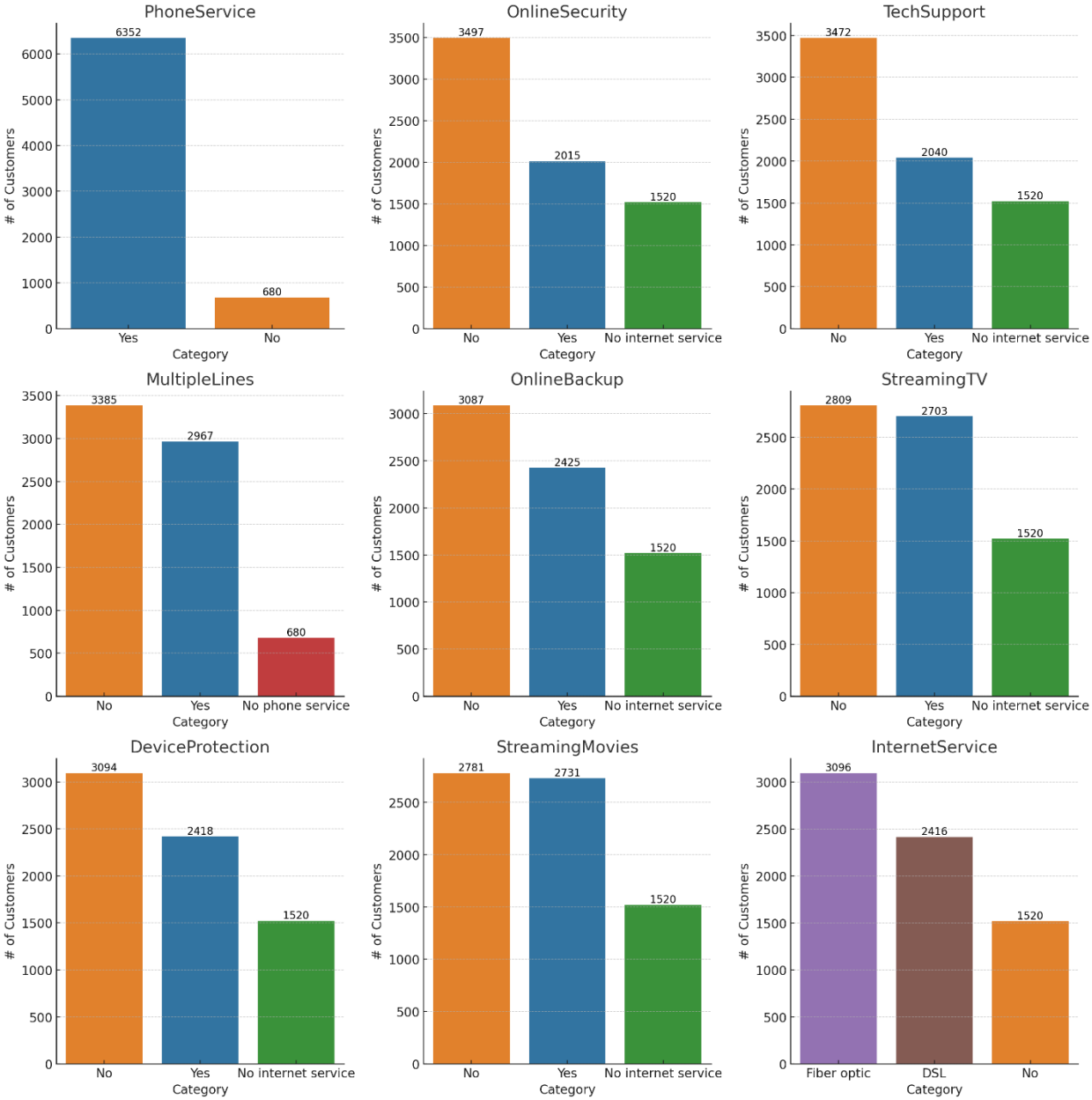


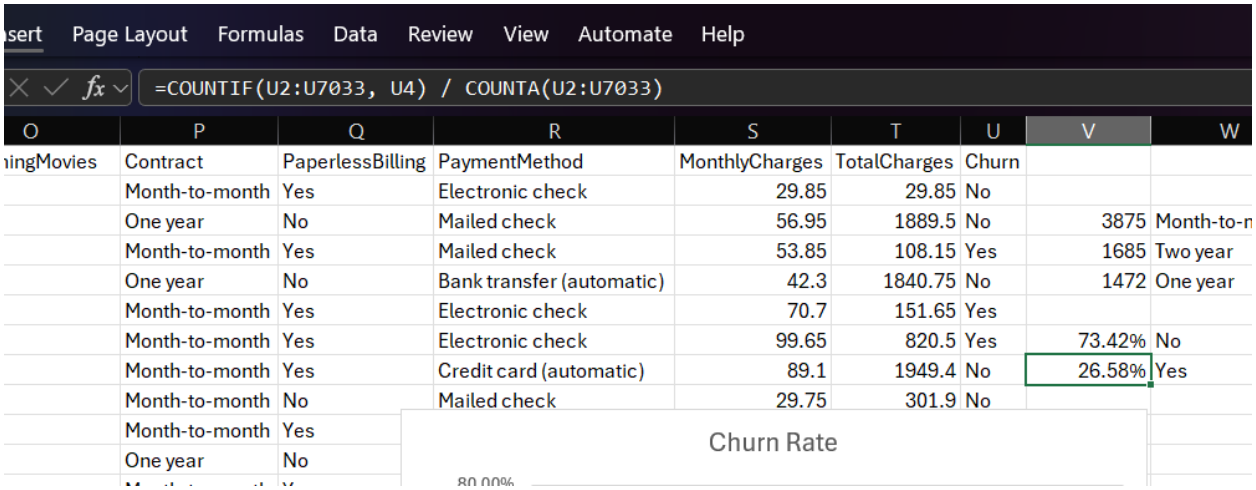
As we can see from this graph most of the customers are in the month to month contract. While there are equal number of customers in the 1 year and 2 year contracts.



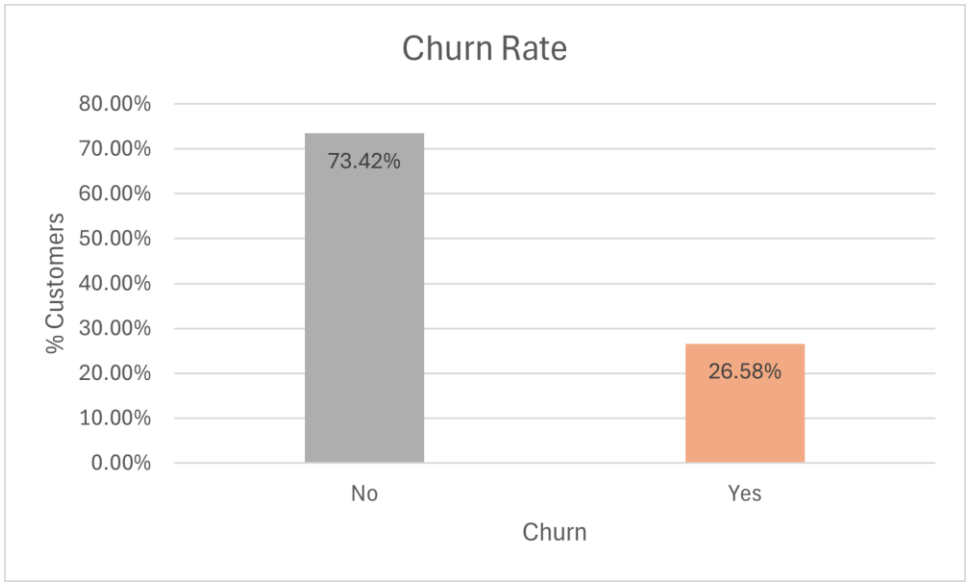
Interestingly most of the monthly contracts last for 1-2 months, while the 2 year contracts tend to last for about 70 months. This shows that the customers taking a longer contract are more loyal to the company and tend to stay with it for a longer period of time.

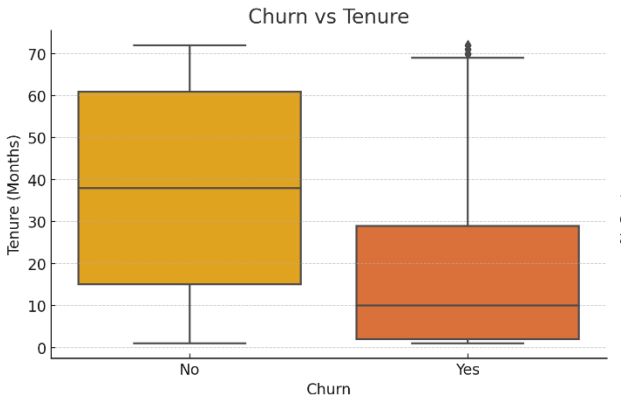
Let us now look at the distribution of various services used by customers¶


PhoneService
6352
680
Yes
No
of Customers
Category
OnlineSecurity
3497
2015
1520
No
Yes
No internet service
TechSupport
3472
2040
1520
MultipleLines
3385
2967
680
No phone service
OnlineBackup
3087
2425
1520
StreamingTV
2809
2703
1520
DeviceProtection
3094
2418
1520
StreamingMovies
2781
2731
1520
InternetService
3096
2416
1520
Fiber optic
DSL

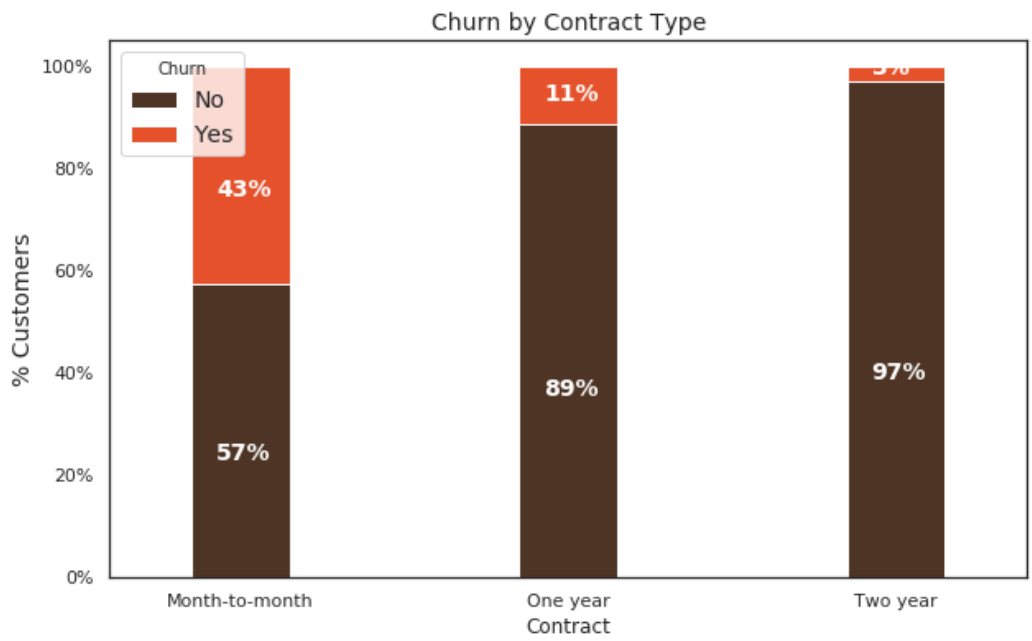


O	P	Q	R	S	T	U	V	W
ingMovies	Contract	PaperlessBilling	PaymentMethod	MonthlyCharges	TotalCharges	Churn		
	Month-to-month	Yes	Electronic check	29.85	29.85	No		
	One year	No	Mailed check	56.95	1889.5	No	3875	Month-to-m
	Month-to-month	Yes	Mailed check	53.85	108.15	Yes	1685	Two year
	One year	No	Bank transfer (automatic)	42.3	1840.75	No	1472	One year
	Month-to-month	Yes	Electronic check	70.7	151.65	Yes		
	Month-to-month	Yes	Electronic check	99.65	820.5	Yes	73.42%	No
	Month-to-month	Yes	Credit card (automatic)	89.1	1949.4	No	26.58%	Yes
	Month-to-month	No	Mailed check	29.75	301.9	No		
	Month-to-month	Yes						
	One year	No						

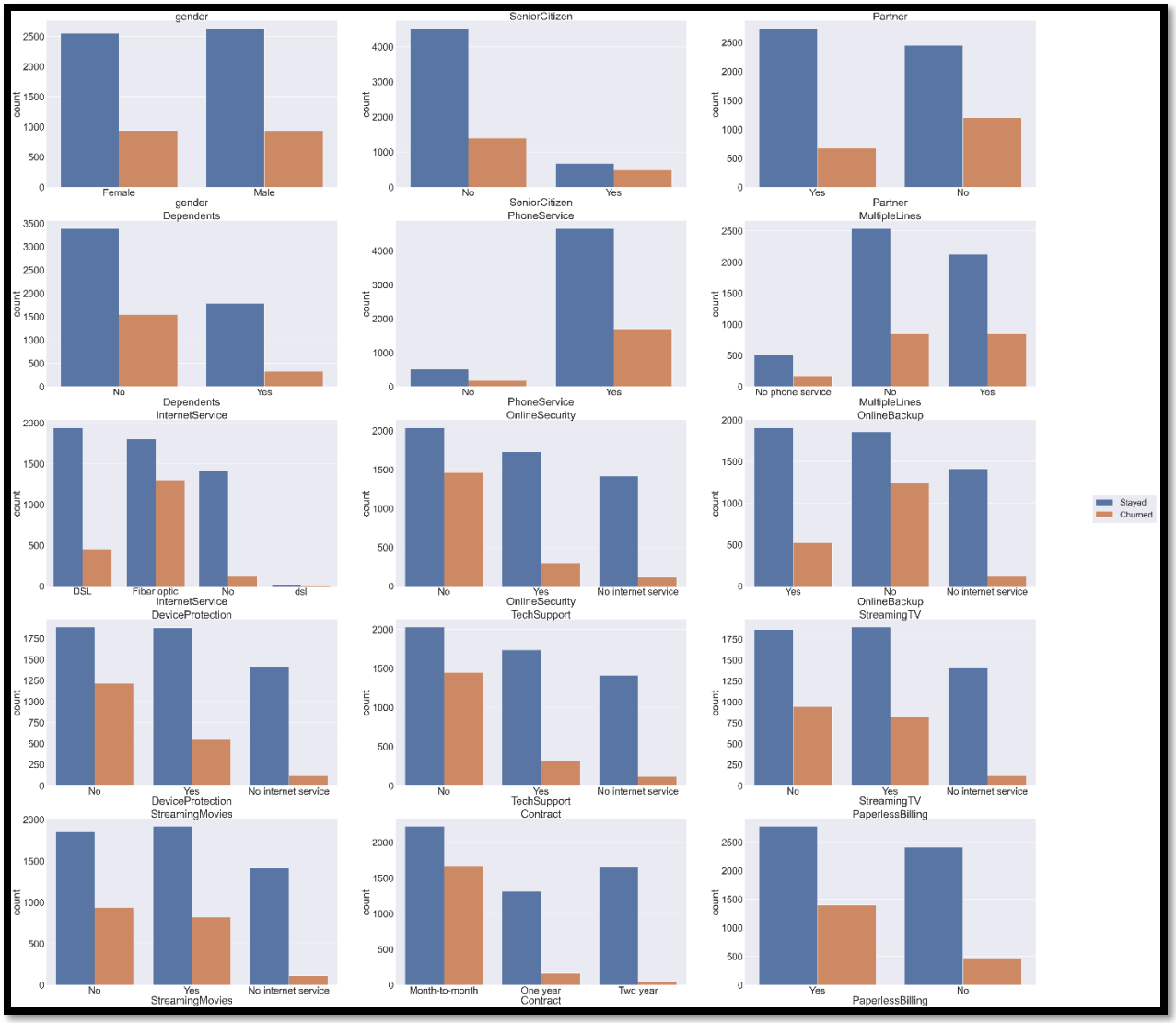


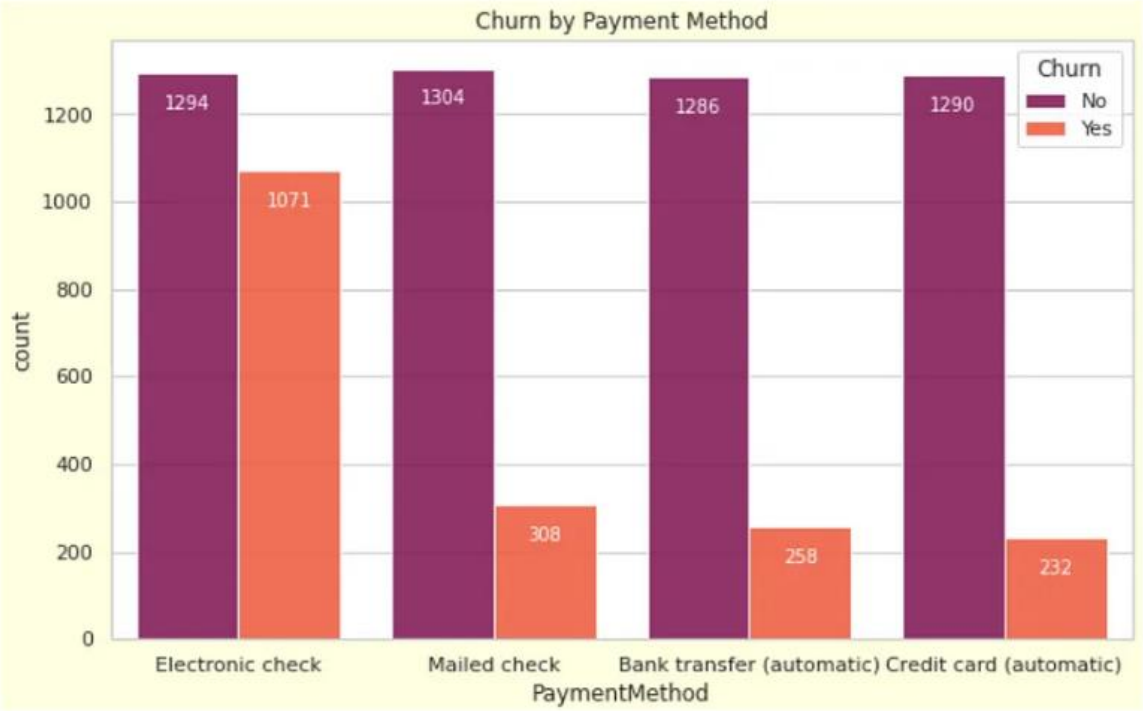


Churn vs Tenure: As we can see form the below plot, the customers who do not churn, they tend to stay for a longer tenure with the telecom company.



the customers who have a month to month contract have a very high churn rate.


gender
SeniorCitizen
Partner
Dependents
PhoneService
MultipleLines
InternetService
OnlineSecurity
OnlineBackup
DeviceProtection
TechSupport
StreamingTV
StreamingMovies
Contract
PaperlessBilling
count
Female
Male
No
Yes
No phone service
No internet service
DSL
Fiber optic
dsl
Month-to-month
One year
Two year
Stayed
Churned

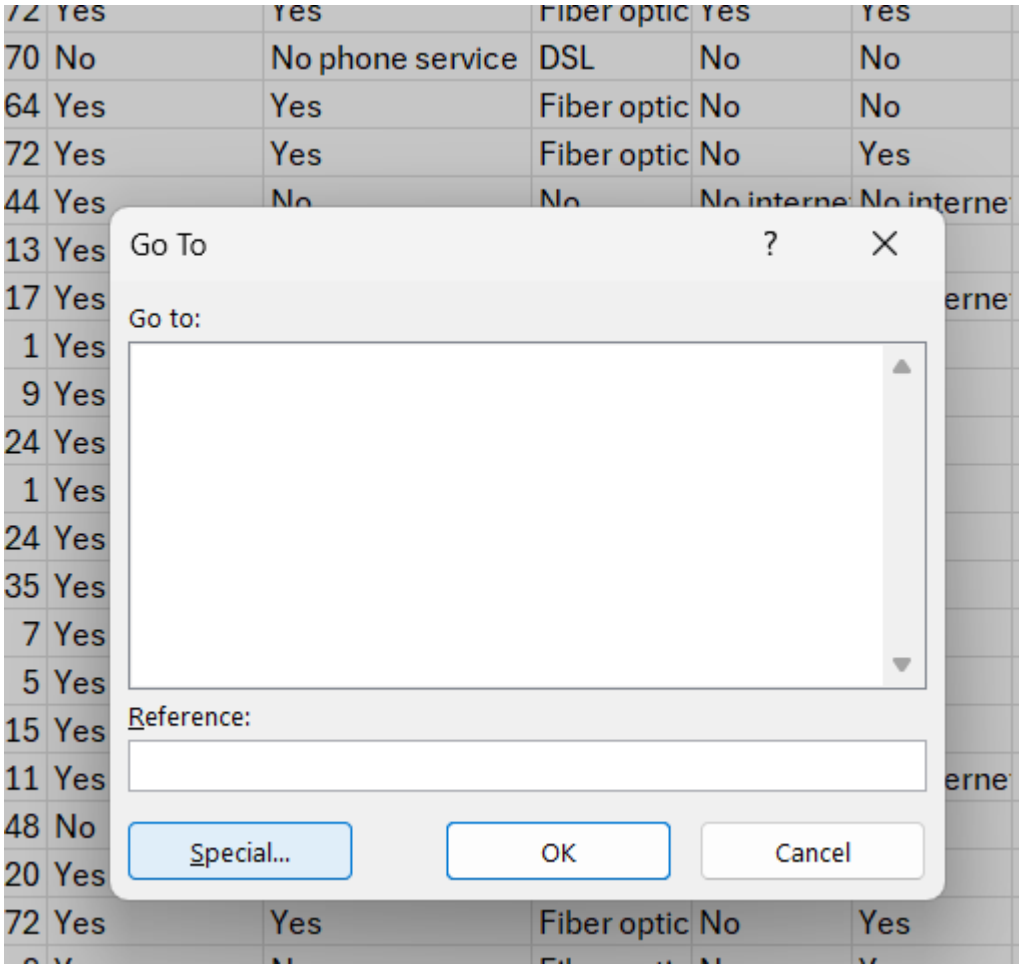


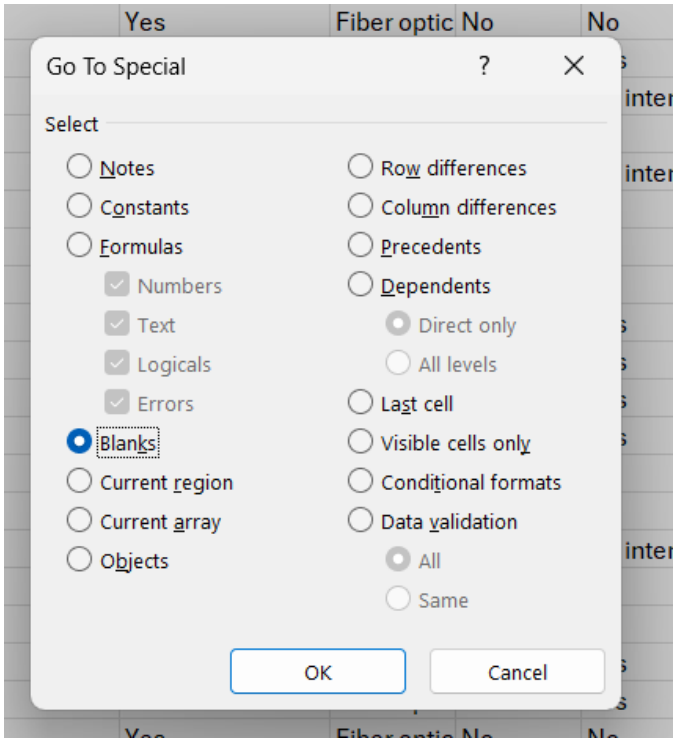
3-Data preparation

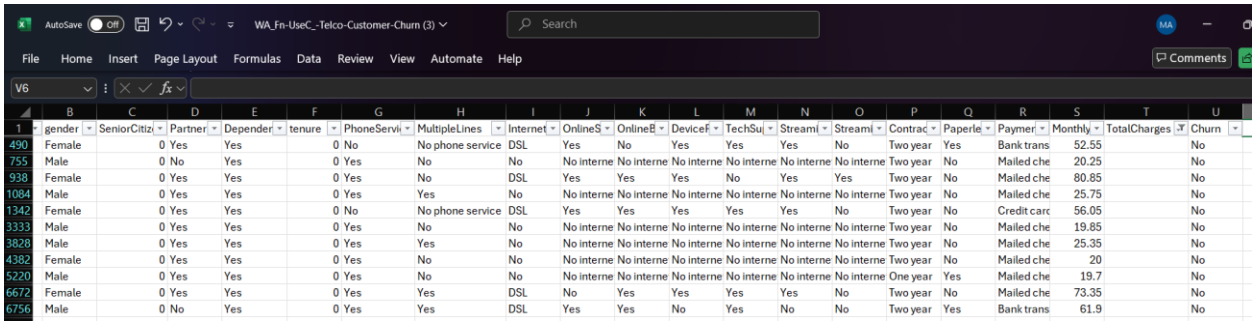
Cleaning & Preprocessing Steps:

- **Removed missing values** from TotalCharges (spotted blanks, then dropped rows).

  In excel, ctrl A then control g

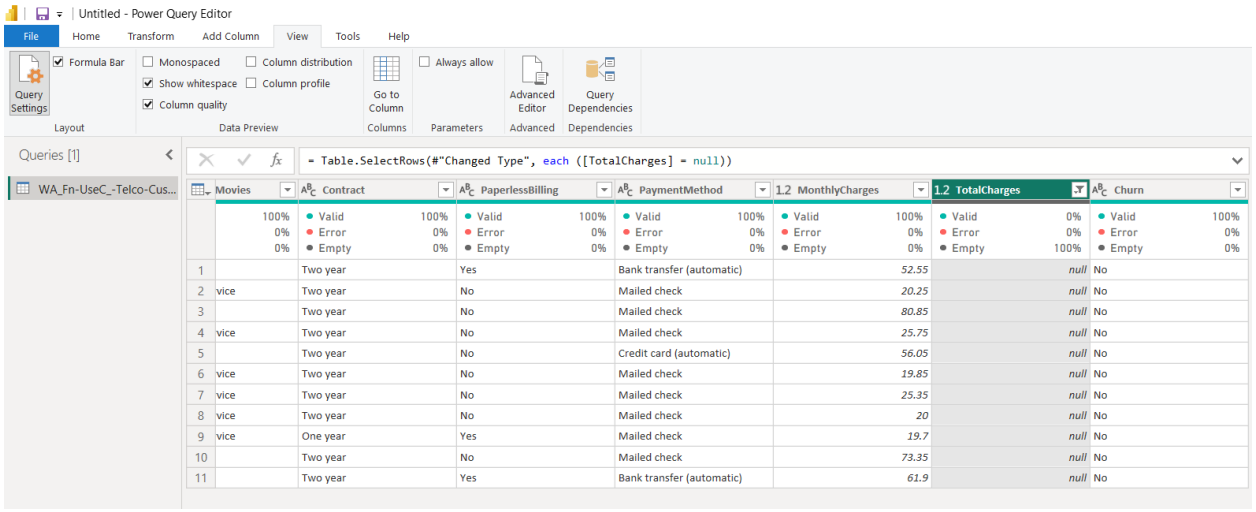

70 No No phone service DSL No No
64 Yes Yes Fiber optic No No
72 Yes Yes Fiber optic No Yes
Go To
Go to:
Reference:
Special...
OK
Cancel
72 Yes Yes Fiber optic No Yes


Yes Fiber optic No No
Go To Special
Select
Notes
Constants
Formulas
Numbers
Text
Logicals
Errors
Blanks
Current region
Current array
Objects
Row differences
Column differences
Precedents
Dependents
Direct only
All levels
Last cell
Visible cells only
Conditional formats
Data validation
All
Same
OK
Cancel



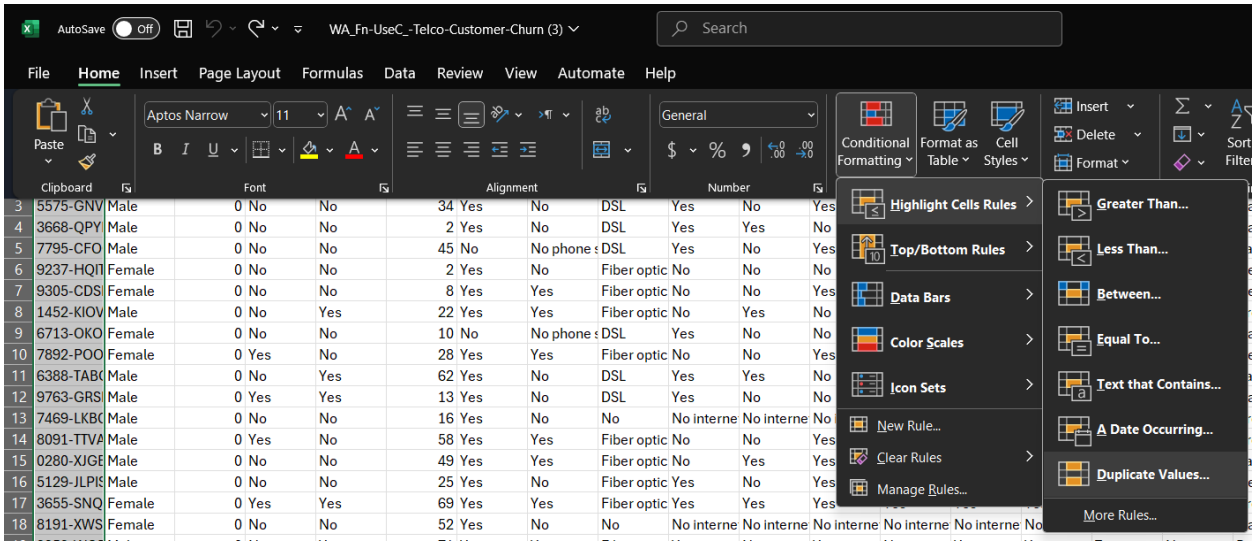
	gender	SeniorCitiz	Partner	Dependen	tenure	PhoneServi	MultipleLines	Internet	OnlineS	OnlineB	DeviceP	TechSu	Streami	Streami	Contrac	Paperle	Paymer	Monthly	TotalCharges	Churn
490	Female	0	Yes	Yes	0	No	No phone service	DSL	Yes	No	Yes	Yes	Yes	No	Two year	Yes	Bank trans	52.55		No
755	Male	0	No	Yes	0	Yes	No	No	No interne	No interne	No interne	No interne	No interne	No interne	Two year	No	Mailed che	20.25		No
938	Female	0	Yes	Yes	0	Yes	No	DSL	Yes	Yes	Yes	No	Yes	Yes	Two year	No	Mailed che	80.85		No
1084	Male	0	Yes	Yes	0	Yes	Yes	No	No interne	No interne	No interne	No interne	No interne	No interne	Two year	No	Mailed che	25.75		No
1342	Female	0	Yes	Yes	0	No	No phone service	DSL	Yes	Yes	Yes	Yes	Yes	No	Two year	No	Credit carc	56.05		No
3333	Male	0	Yes	Yes	0	Yes	No	No	No interne	No interne	No interne	No interne	No interne	No interne	Two year	No	Mailed che	19.85		No
3828	Male	0	Yes	Yes	0	Yes	Yes	No	No interne	No interne	No interne	No interne	No interne	No interne	Two year	No	Mailed che	25.35		No
4382	Female	0	Yes	Yes	0	Yes	No	No	No interne	No interne	No interne	No interne	No interne	No interne	Two year	No	Mailed che	20		No
5220	Male	0	Yes	Yes	0	Yes	No	No	No interne	No interne	No interne	No interne	No interne	No interne	One year	Yes	Mailed che	19.7		No
6672	Female	0	Yes	Yes	0	Yes	Yes	DSL	No	Yes	Yes	Yes	Yes	No	Two year	No	Mailed che	73.35		No
6756	Male	0	No	Yes	0	Yes	Yes	DSL	Yes	Yes	No	Yes	No	No	Two year	Yes	Bank trans	61.9		No

Power bi



	Movies	Contract	PaperlessBilling	PaymentMethod	MonthlyCharges	TotalCharges	Churn
	100% Valid 0% Error 0% Empty	100% Valid 0% Error 0% Empty	100% Valid 0% Error 0% Empty	100% Valid 0% Error 0% Empty	100% Valid 0% Error 0% Empty	0% Valid 0% Error 100% Empty	100% Valid 0% Error 0% Empty
1		Two year	Yes	Bank transfer (automatic)	52.55	null	No
2	vice	Two year	No	Mailed check	20.25	null	No
3		Two year	No	Mailed check	80.85	null	No
4	vice	Two year	No	Mailed check	25.75	null	No
5		Two year	No	Credit card (automatic)	56.05	null	No
6	vice	Two year	No	Mailed check	19.85	null	No
7	vice	Two year	No	Mailed check	25.35	null	No
8	vice	Two year	No	Mailed check	20	null	No
9	vice	One year	Yes	Mailed check	19.7	null	No
10		Two year	No	Mailed check	73.35	null	No
11		Two year	Yes	Bank transfer (automatic)	61.9	null	No

Remove duplicates based on customerID.



3	5575-GNV	Male	0	No	No	34	Yes	No	DSL	Yes	No	Yes							
4	3668-QPY	Male	0	No	No	2	Yes	No	DSL	Yes	Yes	No							
5	7795-CFO	Male	0	No	No	45	No	No phone s	DSL	Yes	No	Yes							
6	9237-HQI	Female	0	No	No	2	Yes	No	Fiber optic	No	No	No							
7	9305-CDS	Female	0	No	No	8	Yes	Yes	Fiber optic	No	No	Yes							
8	1452-KIOV	Male	0	No	Yes	22	Yes	Yes	Fiber optic	No	Yes	No							
9	6713-OKO	Female	0	No	No	10	No	No phone s	DSL	Yes	No	No							
10	7892-POO	Female	0	Yes	No	28	Yes	Yes	Fiber optic	No	No	Yes							
11	6388-TAB	Male	0	No	Yes	62	Yes	No	DSL	Yes	Yes	No							
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13	7469-LKB	Male	0	No	No	16	Yes	No	No	No interne	No interne	No							
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15	0280-XJG	Male	0	No	No	49	Yes	Yes	Fiber optic	No	Yes	Yes							
16	5129-JLPI	Male	0	No	No	25	Yes	No	Fiber optic	Yes	No	Yes							
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7031	2235-DWLJU	Female	1	No	No
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7039	2569-WGERO	Female	0	No	No
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7044	3186-AJIEK	Male	0	No	No
7045	0871-OPBXW	Female	0	No	No
7046	3605-JISKB	Male	1	Yes	No

4-modeling

1.Churn Rate by Contract Type:

- **KPI**: **Churn Rate by Contract Type** (calculated as a percentage)
- **Formula**:

Churn Rate=Churned Customers/Total Customers in Contract Type×100

- **Why it matters**: This helps in assesing how different contract types (e.g., month-to-month, one-year, two-year) influence customer churn. Contract type is a key factor in retention, and this KPI helps target efforts to reduce churn based on contract length.

2.Churn vs. Tenure (Box Plot):

- **KPI**: **Average Tenure** and the **Distribution of Tenure** for churned vs. non-churned customers.
- **Why it matters**: **Tenure** plays a major role in churn behavior. Customers with a longer tenure are generally less likely to churn. This KPI helps understand the length of time a customer typically stays before churning and shows the distribution of tenure values for churned vs. stayed customers.

3. **Average Tenure (Total, Stayed, Churned)**:

- **KPI**: **Average Tenure** for churned vs. stayed customers.
- **Why it matters**: By comparing the average tenure of churned and stayed customers, we can identify the difference in loyalty and how long customers typically remain with the service before churning. It’s important for understanding retention strategies.

4. **Total Charges (Sum of Total, Stayed, Churned)**:

- **KPI**: **Total Revenue from Churned and Stayed Customers**
- **Why it matters**: This KPI helps measure the financial impact of churn. It compares the total revenue generated by churned vs. stayed customers, highlighting the revenue loss from churn and the value retained by loyal customers.
- 5. **Monthly Charges (Sum of Total, Stayed, Churned)**:
- **KPI**: **Monthly Revenue from Churned and Stayed Customers**

- **Why it matters**: This shows how much recurring monthly revenue you are losing or gaining due to churn. It's a crucial KPI for understanding the impact of churn on short-term revenue.

6 **Total Charges Average (Total, Stayed, Churned)**:

- **KPI**: **Average Revenue per Customer**
- **Why it matters**: This helps assess the **average spending** by churned and retained customers, offering insight into customer value. It reveals how much each group contributes to the total revenue.

7 **Monthly Charges Average (Total, Stayed, Churned)**:

- **KPI**: **Average Monthly Revenue per Customer**
- **Why it matters**: Similar to the total charges average, this helps us understand **monthly customer value** and compare churned vs. stayed customers in terms of how much they pay each month.

b. Use BI Tools (Power BI, Tableau, or Excel) to Calculate KPIs

1. **Churn Rate by Contract Type**:
    - Formula:

      =COUNTIFS(P:P, "Month-to-month",U:U, "No") / COUNTIF(P:P, "Month-to-month")

      =COUNTIFS(P:P, "Month-to-month",U:U, "Yes") / COUNTIF(P:P, "Month-to-month")

      =COUNTIFS(P:P, "One year",U:U, "No") / COUNTIF(P:P, "One year")

      =COUNTIFS(P:P, "One year",U:U, "Yes") / COUNTIF(P:P, "One year")

      =COUNTIFS(P:P, "Two year",U:U, "No") / COUNTIF(P:P, "Two year")

      =COUNTIFS(P:P, "Two year",U:U, "Yes") / COUNTIF(P:P, "Two year")

2.**Average Tenure (Total, Stayed, Churned)**:

=AVERAGE(F:F)

=AVERAGEIFS(F:F, U:U, "Yes")

=AVERAGEIFS(F:F, U:U, "No")

3.**Total Charges (Sum of Total, Stayed, Churned)**:

=SUM(T:T)

=SUMIF(U:U, "Yes", T:T)

=SUMIF(U:U, "No", T:T)

4. Monthly Charges (Sum of Total, Stayed, Churned):

=SUM(S2:S7033)

=SUMIF(U2:U7033, "Yes", S2:S7033)

=SUMIF(U2:U7033, "No", S2:S7033)

5. **Total Charges Average (Total, Stayed, Churned)**:

=AVERAGE(T:T)

=AVERAGEIFS(T:T, U:U, "Yes")

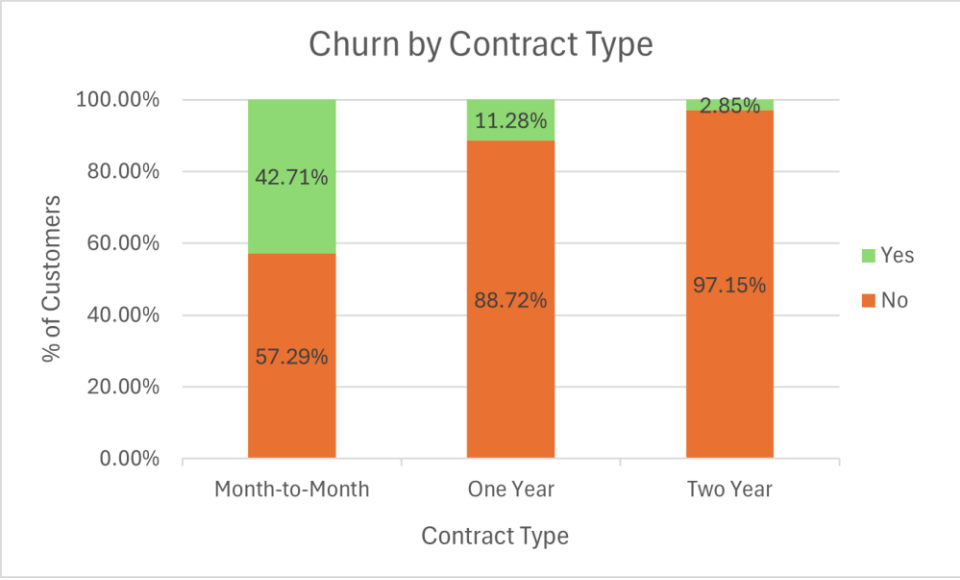
=AVERAGEIFS(T:T, U:U, "No")

6. **Monthly Charges Average (Total, Stayed, Churned)**:

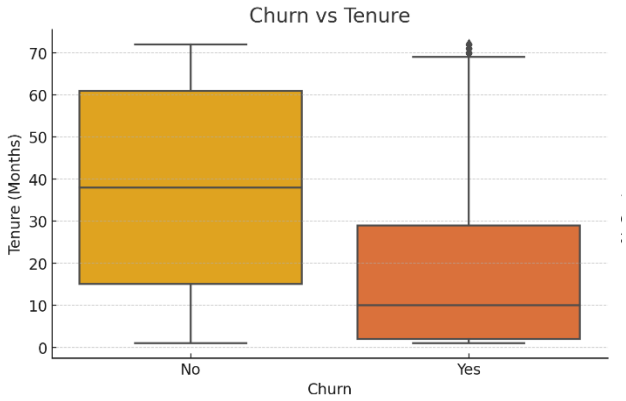
=AVERAGE(S:S)

=AVERAGEIFS(S:S, U:U, "Yes")

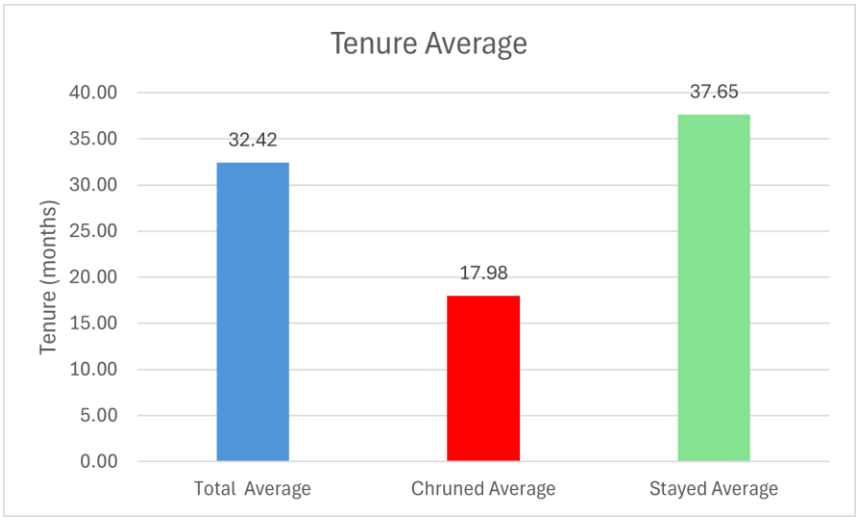
=AVERAGEIFS(S:S, U:U, "No")



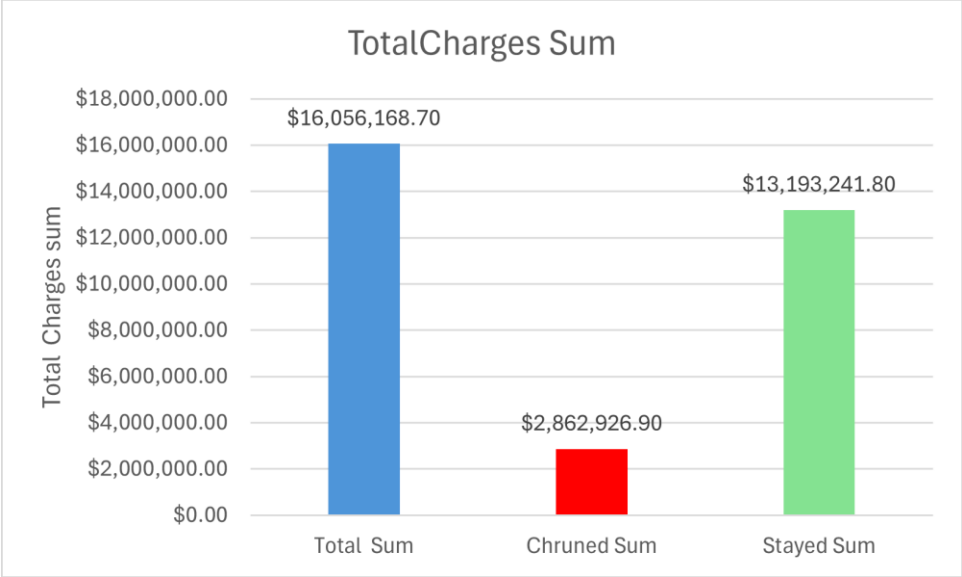
This stacked bar chart visualizes the churn rate by contract type, with the percentage of churned vs. retained customers for month-to-month, one-year, and two-year contracts. It shows that the month-to-month contract type has the highest churn rate, with over 42% of customers churning, while the two-year contract has the lowest churn rate at just 2.85%.



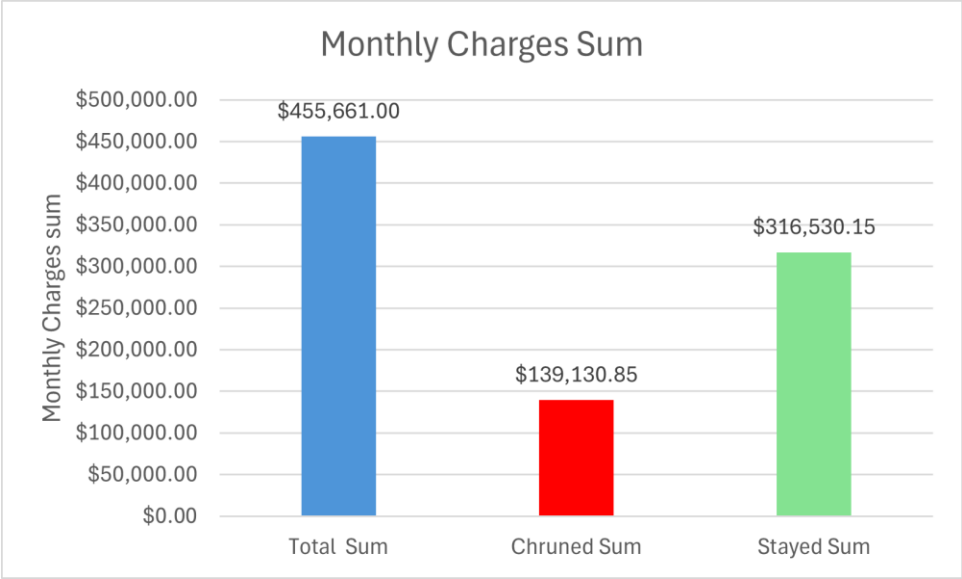
This box plot illustrates the distribution of tenure (in months) for churned vs. non-churned customers. The chart highlights that customers who stayed tend to have significantly higher tenure, with most churned customers having lower tenure values.



This chart visualizes the average tenure (in months) of customers in total, churned, and stayed categories. It shows that customers who stayed have a significantly higher average tenure, reflecting stronger retention among loyal customers.

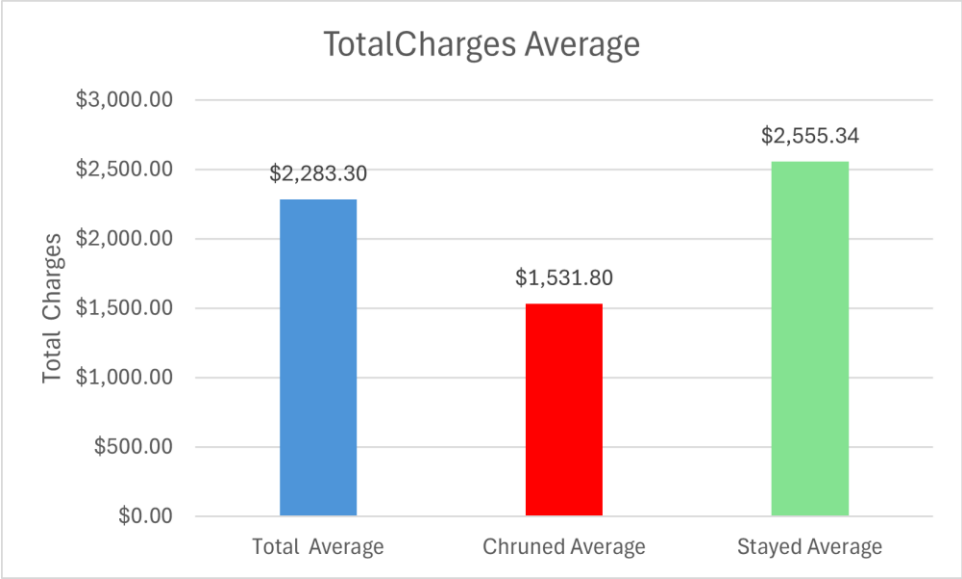


This bar chart represents the total revenue generated from customers in three categories: total, stayed, and churned. It shows that the total revenue generated by customers who stayed is significantly higher compared to churned customers, indicating a financial impact of churn on revenue.

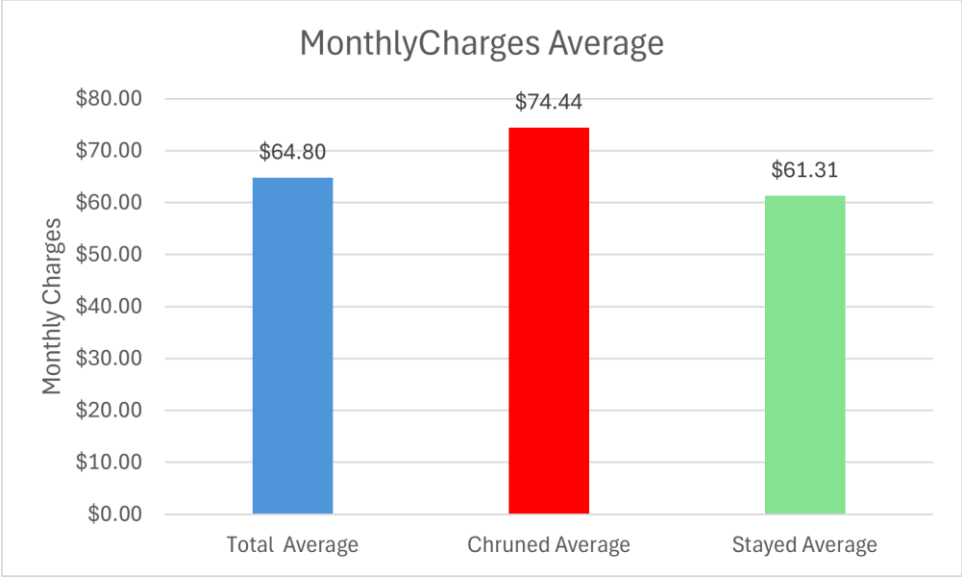


This chart compares the total, churned, and stayed customers based on their monthly charges. It indicates that the stayed customers generate a larger portion of

the monthly revenue compared to churned customers.



This bar chart shows the average total charges per customer, broken down into total, churned, and stayed categories. It reveals that customers who stayed have a higher average total charge compared to churned customers, suggesting a greater customer lifetime value for retained users.



This bar chart compares the average monthly charges for customers in total, churned, and stayed groups. Customers who stayed have higher monthly charges on average compared to churned customers, indicating that loyal customers contribute more to monthly revenue.

5. Evaluation

a. Interpret the Results from the KPIs and Visualizations

1. Churn Rate by Contract Type:

- **Interpretation**: The stacked bar chart shows that **Month-to-month** contract customers exhibit the highest churn rate (42.71%), while **Two-year** contracts have the lowest churn rate (2.85%). This suggests that customers on longer-term contracts (e.g., two years) are more likely to stay, likely due to a **longer commitment**. The **Month-to-month** customers are more likely to churn because they have the flexibility to leave at any time.

2. Churn vs. Tenure (Box Plot):

- **Interpretation**: The box plot reveals that **churned customers** tend to have a **shorter tenure** with the company, mostly under a year, while **non-churned** customers have significantly longer tenures. The **median tenure** for churned customers is far lower than that for stayed customers, supporting the idea that longer customer retention correlates with reduced churn.

3. Average Tenure (Total, Stayed, Churned):

- **Interpretation**: This bar chart shows that **churned customers** have an average tenure of **around 16 months**, while **stayed customers** average **about 33 months**. This reinforces the finding that longer customer relationships contribute to greater loyalty. The average tenure of churned customers is lower, pointing out the importance of improving retention efforts, especially for newer customers.

4. Total Charges Sum (Total, Stayed, Churned):

- **Interpretation**: The total revenue from **stayed customers** is significantly higher compared to **churned customers**, indicating that **retained customers contribute more significantly to total revenue**. This suggests that focusing on retention would have a substantial impact on the business's overall revenue.

5. Monthly Charges Sum (Total, Stayed, Churned):

- **Interpretation**: The chart shows that **stayed customers** are generating much higher monthly revenue compared to churned customers. This underscores the **long-term value of retained customers** and the importance of minimizing churn to maintain a steady monthly revenue stream.

6. Total Charges Average (Total, Stayed, Churned):

- **Interpretation**: Customers who stayed with the company tend to have **higher average total charges**, meaning they contribute more revenue over time. The churned customers have a much lower average charge, indicating that **higher-paying customers are more likely to stay**. This highlights the financial benefits of retaining higher-value customers.

7. Monthly Charges Average (Total, Stayed, Churned):

- **Interpretation**: The average monthly charges for **stayed customers** are **higher** compared to churned customers. This suggests that **long-term customers** are willing to spend more on a monthly basis, likely due to upselling, added services, or loyalty over time.

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b. Assess How Well the Results Meet the Business Objectives Defined Earlier

The business objectives outlined earlier likely focus on **reducing churn**, **improving customer retention**, and **maximizing revenue**. Here's how your KPIs meet these objectives:

1. **Reducing Churn**:
   The **Churn Rate by Contract Type** and **Churn vs. Tenure** KPIs show that **month-to-month contracts** have the highest churn, suggesting a target area for retention efforts. By focusing on reducing churn in **month-to-month contracts**, the company can significantly improve retention.
2. **Improving Customer Retention**:
   The **Average Tenure** and **Churn vs. Tenure** visualizations support the idea that longer-tenure customers are more loyal and less likely to churn. This suggests that investing in **loyalty programs** or providing incentives for longer-term contracts could be an effective strategy.
3. **Maximizing Revenue**:
   The **Total Charges Sum**, **Monthly Charges Sum**, **Total Charges Average**, and **Monthly Charges Average** all indicate that **stayed customers contribute more significantly to revenue**. Focusing on retention, particularly for high-paying customers, will likely **maximize revenue**. The lower charges from churned customers highlight a **loss in revenue** that could be recovered by improving retention strategies.

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c. Highlight Trends, Patterns, or Anomalies and Their Implications for the Business Problem

1. High Churn in Month-to-Month Contracts:

- **Trend/Pattern**: The **high churn rate** in **month-to-month contracts** is a significant finding. This contract type has a higher churn rate compared to others.

- **Implications**: The company might consider offering incentives, such as **discounts or rewards** for longer commitments, or **upgrading month-to-month contracts** to **one- or two-year contracts**. Additionally, addressing **customer satisfaction issues** for month-to-month customers might help lower churn.

2. Longer Tenure Correlates with Lower Churn:

- **Trend/Pattern**: Customers who have been with the company longer are much more likely to stay. The **box plot** clearly shows that churned customers have shorter tenure.
- **Implications**: **Customer retention strategies** should focus on keeping new customers engaged for longer. Providing **onboarding programs**, personalized offers, or **loyalty rewards** early in the customer lifecycle could reduce churn and increase retention.

3. Retained Customers Contribute More to Revenue:

- **Trend/Pattern**: The **monthly charges** and **total charges** for stayed customers are significantly higher than churned customers.
- **Implications**: **Retention is key to increasing long-term revenue**. Investing in strategies to retain higher-value customers (e.g., **premium plans or upsell opportunities**) could significantly impact the company's overall financial health. The company might also focus on improving the **customer experience** for high-revenue customers to keep them loyal.

4. Financial Impact of Churn:

- **Trend/Pattern**: The **revenue from churned customers** is much lower than that from stayed customers, highlighting the financial impact of churn.
- **Implications**: **Reducing churn** is essential to maintaining a stable revenue stream. The company might explore strategies such as **targeted marketing**, **customer feedback surveys**, and **customer support interventions** to identify and address reasons for churn before it happens.

6.Deployment

a. Provide At Least Three Actionable Recommendations Based on Your Findings

1. **Recommendation 1: Incentivize Long-Term Contracts**
    - **Finding**: Customers on **month-to-month contracts** have a **much higher churn rate** compared to those on **one-year** or **two-year contracts**.
    - **Action**: Offer **discounts or loyalty rewards** for customers who switch from month-to-month contracts to longer-term contracts (1 or 2 years).
    - **Objective**: To reduce churn, particularly in the month-to-month contract group, and increase customer loyalty. Customers who commit to longer contracts are less likely to churn.
2. **Recommendation 2: Focus on Onboarding and Early Engagement for New Customers**
    - **Finding**: **Churned customers** tend to have **shorter tenure**, indicating that newer customers are more likely to leave early.
    - **Action**: Implement a comprehensive **onboarding program** that provides clear value propositions, customer education, and early-stage support to ensure customers engage with the service and are less likely to churn in their first few months.
    - **Objective**: To reduce early-stage churn by improving engagement with new customers and making them feel valued and informed from the start.
3. **Recommendation 3: Target High-Value Customers with Premium Offers and Upselling**
    - **Finding**: **Stayed customers** tend to generate **higher total and monthly charges** than churned customers, showing that loyal customers contribute more to the company's revenue.
    - **Action**: **Upsell** and **cross-sell** higher-value plans or additional services to customers who have been with the company for an extended period. Offer **premium customer support** or exclusive deals to incentivize continued loyalty.
    - **Objective**: To increase **average revenue per user (ARPU)** and maximize the lifetime value of loyal customers.

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b. Discuss How These Recommendations Could Be Implemented in a Real-World Scenario

1. Incentivize Long-Term Contracts

- **Implementation**:
    - **Offer Discounts**: Provide **significant discounts** for customers who switch from a month-to-month contract to a one- or two-year contract. For example, offer a **5-10% discount** for customers who upgrade to longer contracts.
    - **Create Loyalty Programs**: Introduce a **loyalty program** that rewards customers for signing long-term contracts, with perks such as **free months of service**, **exclusive content**, or **priority customer support**.
    - **Communicate Value**: Highlight the **cost savings** and **predictable service** that come with longer-term contracts in marketing materials, emails, and customer service communications.
    - **Real-World Example**: A telecom company could send targeted email offers to month-to-month customers, offering them a **free month** if they commit to a **one-year plan**. The company could also send push notifications through its mobile app, emphasizing the benefits of switching.

2. Focus on Onboarding and Early Engagement for New Customers

- **Implementation**:
    - **Personalized Onboarding**: Create a **personalized onboarding journey** that includes clear, step-by-step instructions on how to get the most out of the service. Offer **live chat support** or **self-help tutorials** in the first few months.
    - **Customer Success Managers**: Assign **customer success managers** (or support agents) to **new customers** to ensure that their first few interactions are positive and they receive any needed support promptly.
    - **Engagement Campaigns**: Launch **email or SMS campaigns** designed to keep customers engaged during the early months. Provide value-driven content, such as **how-to guides**, **service tips**, and **rewards for completing certain milestones** (e.g., using the app, paying bills on time).
    - **Real-World Example**: A telecom company could implement an automated email sequence for new customers that offers them a **welcome discount**

on their second bill, guides them through the top features of their plan, and reminds them to use customer support if needed.

3. Target High-Value Customers with Premium Offers and Upselling

- **Implementation**:
    - **Identify High-Value Customers**: Use **data analysis** to identify customers with a **high lifetime value (LTV)**, typically those who have been with the company for a long time and have higher-than-average monthly charges.
    - **Offer Exclusive Premium Plans**: Create **premium plans** or **upsell opportunities** for loyal customers that provide **extra features**, **better customer support**, or **exclusive content**.
    - **Personalized Deals**: Send **personalized offers** to high-value customers based on their usage patterns and preferences. For example, offer a **higher-tier plan** at a discounted rate or provide **add-ons** like additional mobile data, streaming services, or international roaming.
    - **Real-World Example**: A telecom provider could analyze its existing customers and identify those who are paying for mid-tier plans but may benefit from a **higher-tier plan**. The provider could then send these customers personalized offers through email or the company's app, offering a **discounted upgrade** or **free trial of premium services**.

Recommendation	Action Plan	Timeline	Resources Needed
Incentivize Long-Term Contracts	1. Offer discounts for switching to long-term contracts. 2. Introduce loyalty programs. 3. Market contract benefits via email and app notifications.	1-3 months	Marketing team, Discount structures, CRM software
Focus on Onboarding and Early Engagement	1. Create personalized onboarding journeys. 2. Provide customer success managers. 3. Launch early-stage engagement campaigns.	1-2 months	Customer support team, Email marketing tools, CRM

Recommendation	Action Plan	Timeline	Resources Needed
Target High-Value Customers with Premium Offers	1. Analyze data to identify high-value customers. 2. Offer premium plans and personalized deals.	2-4 months	Data analysis tools, Sales team, Customer insights